

Predetermined Change Control Plan (AID algorithm)

Grounded in: Insulet Omnipod 5 (AID algorithm)

Basis: FDA PCCP final guidance (2024).

Device	Insulet Omnipod 5 — AID dosing algorithm
Status	First FDA-cleared tubeless AID; algorithm enhancement 510(k) (Dec 2025)
Basis	FDA PCCP final guidance (2024). Illustrative — algorithm change control.

1. Description of modifications

- Bounded changes to target-glucose options and control parameters (intended use unchanged).
- Adding compatibility with additional cleared CGM sensors after equivalence testing.

2. Modification protocol

Pre-specified safety bounds (max dose, target ranges); in-silico + clinical validation; acceptance gates on time-in-range and safety (no increase in hypo events); versioned release with rollback.

3. Impact assessment

Modification	Risk	Control / verification
Algorithm parameter change	Hypo-/hyperglycemia	Safety bounds + clinical acceptance gates
New CGM compatibility	Input domain shift	Sensor equivalence testing before enablement

4. Transparency

Labeling version notes per in-scope release; changes beyond the envelope require a new submission.

Sources (public)

- Insulet — Omnipod 5, first FDA-cleared tubeless automated insulin delivery (AID) patch pump; smartphone-controlled.
- Public facts: integrated dosing algorithm adjusts insulin every 5 min using CGM input; 510(k) algorithm update (Dec 2025).
- FDA — Premarket Cybersecurity guidance (2025); PCCP final guidance (2024); GMLP.